LitTraceQA: A Benchmark for Multi-Stage Grounding and Verification in Scientific Question Answering

Xuye Liu[1] **Yimu Wang**[1] **Peng Shi**[1] **Bo Xue**[2] **Xiangrui Ke**[1]
Songcheng Cai[1] **Kath Choi**[1] **Di Wu**[3] **Freda Shi**[1] **Krzysztof Czarnecki**[1]
[1]University of Waterloo [2]City University of Hong Kong [3]University of Amsterdam

Abstract

Scientific question answering is useful only when readers can recover the papers and evidence behind an answer. Yet many benchmarks either provide the source document or score only the final response, obscuring failures in literature retrieval and evidence localization. We present LITTRACEQA, a literature-grounded question answering benchmark in which a system must return a connected trace: canonical paper identifiers, typed evidence locations, and a checkable answer. The collection contains 4,978 unique questions grounded in 4,859 papers and covers tables, figures, text spans, equations or algorithms, and citation contexts. Of these questions, 64.85% require multiple papers; answers use multiple-choice or structured-table formats; and 61.9% are missed by every memory-only challenger used during construction. We describe a generate–ground–challenge pipeline and an evaluation protocol that reports paper retrieval, evidence grounding, answer correctness, and strict joint success separately. LitTraceQA therefore tests whether scientific QA systems can produce answers whose provenance is inspectable, rather than merely plausible.

1 Introduction

Language models increasingly serve as interfaces to scientific literature, including retrieval-augmented assistants and literature-synthesis systems (Lewis et al., 2020; Lála et al., 2023; Asai et al., 2024). A scientifically useful answer, however, is not defined by fluency alone. A reader should be able to identify the paper that supports each claim, locate the relevant table, figure, passage, equation, algorithm, or citation context, and verify that the reported answer follows from that evidence. This requirement is especially important for comparative questions, where the answer may depend on results or definitions distributed across several papers.

Existing scientific QA resources cover important parts of this process. QASPER and QASA study evidence-based reasoning over supplied papers (Dasigi et al., 2021; Lee et al., 2023); SPIQA and SCITAT emphasize scientific figures, tables, and multimodal reasoning (Pramanick et al., 2024; Zhang et al., 2025); M3SciQA considers multi-document scientific QA (Li et al., 2024); and LitSearch evaluates realistic literature retrieval (Ajith et al., 2024). Attribution and verification benchmarks likewise show that an answer can sound correct while its cited support is incomplete or irrelevant (Wadden et al., 2020; Bohnet et al., 2022; Gao et al., 2023). What remains less directly measured is the complete trace from a broad paper pool to the precise evidence and then to a checkable answer.

We introduce LITTRACEQA, which treats this trace as the prediction target. Given a question and paper metadata pool, a system returns (i) relevant paper identifiers, (ii) typed evidence locations inside those papers, and (iii) the requested answer components. The design exposes three distinct failure modes: retrieving the wrong paper, grounding the right paper at the wrong location, and deriving the wrong answer from correct evidence. It also supports both hidden-source single-paper questions and multi-paper questions that require evidence alignment rather than independent sub-question solving.

Figure 1 gives the end-to-end view. Questions are constructed from inspectable paper content, grounded against declared sources, challenged without paper access, and retained with their full provenance. Five primary evidence categories reflect common scientific reading behavior, while multiple-choice and structured-table outputs make the resulting answers directly checkable.

Our contributions are threefold. First, we make papers, evidence, and answers jointly required QA outputs. Second, we audit 4,978 unique ques-

Proceedings of the 1st Workshop on Grounding Language Models: Learning Faithfully and Efficiently (GroundLM 2026), pages 1146–1153
October 29, 2026 ©2026 Association for Computational Linguistics

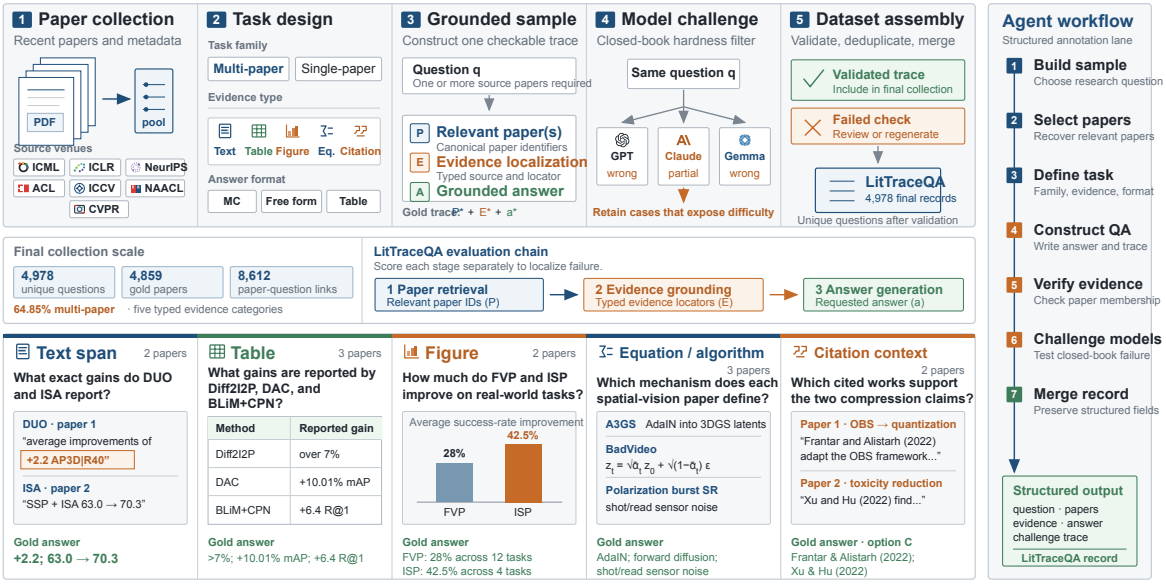


Figure 1: LITTRACEQA connects scientific-paper collection, grounded question construction, closed-book challenge, and validation. Each retained record links a question to one or more gold papers, typed evidence, and a checkable answer; evaluation follows the paper-retrieval, evidence-grounding, and answer-generation trace.

tions spanning five evidence types, including 3,228 multi-paper questions. Third, we specify component and strict joint metrics that expose retrieval, grounding, and answer failures. A 55-example public development set provides the complete release schema for local validation.

2 Task and Collection

2.1 Traceable scientific QA

Let $\mathcal{P}$ be the paper metadata pool and q a research question. A system predicts a paper set $\hat{P} \subseteq \mathcal{P}$, a set of typed evidence items $\hat{E}$, and an answer $\hat{a}$:

$$\hat{P} = \{\hat{p}_1, \ldots, \hat{p}_m\}, \quad (1)$$

$$\hat{E} = \{\hat{e}_1, \ldots, \hat{e}_n\}, \quad (2)$$

$$\hat{a} = \text{Answer}(q, \hat{P}, \hat{E}). \quad (3)$$

The corresponding gold record contains P^*, E^*, and a^*. Canonical paper IDs make retrieval outputs comparable across systems. An evidence item links a paper ID and source type to a locator such as page plus table or figure ID; the annotation can additionally preserve row, column, section, region, equation, algorithm-step, or cited-paper fields. Answers follow the requested format: a discrete option, a structured table, or, in the public development schema, a short free-form component.

The task has three orthogonal axes. **Question scope** distinguishes hidden-source single-paper questions from multi-paper questions. The latter must combine related evidence across papers, for example by aligning methods, datasets, metrics, or cited work. **Evidence type** identifies the primary support as a table, figure, text span, equation or algorithm, or citation context. **Answer format** separates multiple-choice decisions from structured tables; table row keys make cross-paper comparisons machine checkable. A prediction is fully correct only when its papers, evidence, and answer are all correct.

2.2 Generate–ground–challenge construction

The collection is built in four stages. **Source selection** begins from paper metadata and requires locally inspectable text before a paper can support a candidate. **Open-book generation** creates a self-contained question, gold paper set, typed evidence, and answer while the source text is available. Target categories rotate across the five evidence types. **Grounding audit** checks that every evidence item belongs to a declared gold paper and that distinctive text or values can be recovered from the source. **Closed-book challenge** presents the question without papers to several challengers. These attempts do not serve as retrieval-augmented baselines; instead, they record whether a question is likely answerable from model memory alone and expose plausible but unsupported errors. Exact repeated question texts from cumulative checkpoints are collapsed, leaving one record for every unique question.

a Final collection statistics

Statistic	Count	Share
Collection scale		
Unique questions	4,978	–
Unique gold papers	4,859	–
Gold-paper links	8,612	–
Mean papers / question	1.73	–
Mean evidence / question	2.62	–
Question scope		
Multi-paper questions	3,228	64.85%
Single-paper questions	1,750	35.15%
Answer format		
Multiple choice	3,029	60.8%
Structured table	1,949	39.2%
Primary evidence		
Table	1,339	26.9%
Figure	1,132	22.7%
Text span	933	18.7%
Equation / algorithm	819	16.5%
Citation context	755	15.2%
Closed-book hardness		
All challengers wrong	3,080	61.9%
One challenger correct	1,898	38.1%

b

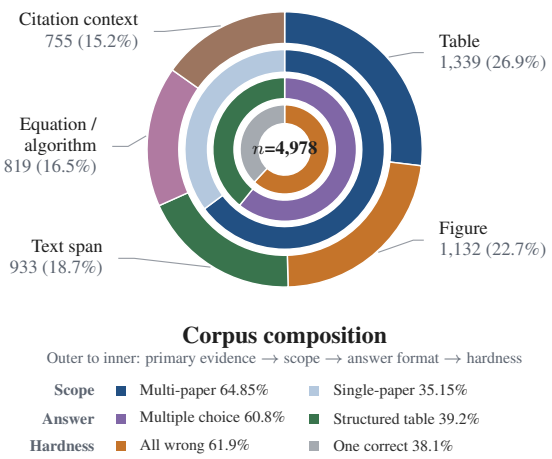


Figure 2: Collection scale and composition after exact-question deduplication. The table gives exact counts; the concentric chart summarizes primary evidence, question scope, answer format, and construction-time closed-book hardness. Hardness is metadata from memory-only challenges, not retrieval-augmented benchmark performance.

2.3 Corpus profile

Figure 2 summarizes the resulting 4,978-question collection. The records reference 4,859 unique gold papers through 8,612 paper links, with 1.73 papers and 2.62 evidence items per question on average. Multi-paper questions account for 3,228 records (64.85%); the remaining 1,750 records provide a diagnostic single-paper setting in which retrieval and internal grounding are still required.

Coverage is deliberately heterogeneous. Primary evidence comprises 1,339 table questions (26.9%), 1,132 figure questions (22.7%), 933 text-span questions (18.7%), 819 equation-or-algorithm questions (16.5%), and 755 citation-context questions (15.2%). Individual records may contain supporting items of several types: across all evidence items, text spans account for 43.5%, tables 16.1%, citation contexts 14.6%, equations or algorithms 13.2%, and figures 12.6%. This distinction between a question's primary type and all supporting sources explains why the two distributions need not match.

Answers are 3,029 multiple-choice records (60.8%) and 1,949 structured tables (39.2%). All memory-only challengers answer incorrectly on 3,080 questions (61.9%); exactly one challenger succeeds on the other 1,898 (38.1%). This filter does not prove retrieval dependence, but it reduces questions that are consistently easy to answer without consulting papers.

3 Evaluation and Analysis

Separated scoring. Paper retrieval is scored over canonical ID sets using question-level precision, recall, and F1, with Recall@K available for ranked outputs. Evidence grounding compares normalized tuples containing paper ID, source type, page, and an object ID when available; exact match and set F1 expose both locator precision and missing support. Answer metrics follow the output type: accuracy for multiple choice and row F1 plus cell accuracy for structured tables. Short free-form components in the public development schema use exact match. **Strict joint success** is a conjunction, not an average: all required paper, evidence, and answer components must be correct.

Diagnostic ladder. End-to-end evaluation supplies only the question and paper pool. An *oracle-paper* setting supplies P^* and isolates evidence localization plus answering; an *oracle-evidence* setting supplies E^* and isolates answer generation. The gap between adjacent settings identifies whether paper retrieval or evidence grounding is the dominant bottleneck. This protocol avoids treating an unsupported correct answer as equivalent to a fully traceable one.

What the audit reveals. The corpus mostly targets multi-paper retrieval, but its 35.15% single-paper subset should be reported separately when claims concern cross-paper reasoning. Its evidence distribution also shows that paragraph extraction is

insufficient: 56.5% of evidence items come from tables, figures, equations or algorithms, and citation contexts. Finally, the closed-book outcomes support the intended difficulty target while remaining construction metadata rather than a model leaderboard. Official retrieval-augmented results are required before making comparative claims about systems.

Representative failure modes. The retained challenge attempts make the benchmark's difficulty more concrete. In one structured-table example, the question asks for both the real-world success-rate improvement and task count reported by FVP and Image-to-Sphere Policy. The gold trace draws on figure context and prose from both papers; the answer is 28% over 12 tasks for FVP and 42.5% over 4 tasks for Image-to-Sphere Policy. All three memory-only challengers filled only one of five checked cells correctly. Their values were plausible in isolation but did not preserve the paper–metric alignment. Another example asks which cited works support an OBS-to-quantization adaptation and monotonic toxicity reduction under knowledge distillation. Every challenger chose a distractor that named related compression work but broke the claim–citation link. These errors motivate trace scoring: answer-only inspection cannot tell a failed cross-paper alignment from a failed citation attribution. Additional examples, including the original error patterns, appear in Appendix E.

Quality-control boundary. The current checks establish structural consistency: evidence belongs to a declared paper, grounding strings are recoverable, answers follow a checkable schema, duplicate question text is removed, and challenge attempts are retained. They do not by themselves establish expert-level correctness. A release-quality benchmark should add stratified human review, report correction rates and inter-annotator agreement, and freeze locator normalization before scoring hidden submissions. Single- and multi-paper subsets should receive separate scores so that paper-internal cases cannot mask weak cross-paper performance.

Intended use. LitTraceQA is designed to diagnose systems that search a bounded scientific corpus and return inspectable support. It should not be interpreted as a test of unrestricted scientific expertise, nor should a traceable citation be taken as proof that a source claim is true. Metadata and annotations can be released, but paper access must remain subject to publisher terms. The dataset card should disclose model-assisted construction, human verification, venue and year concentration, known annotation errors, and the distinction between construction-time challengers and retrieval-augmented evaluation.

4 Related Work

LitTraceQA connects three lines of research. Scientific-document QA provides evidence annotations or multimodal artifacts, but commonly supplies the source paper (Dasigi et al., 2021; Lee et al., 2023; Pramanick et al., 2024; Zhang et al., 2025). Multi-document and scholarly QA expands the source set and synthesis demand (Li et al., 2024; Huang et al., 2025; Yifei et al., 2025), while literature search focuses on paper discovery (Ajith et al., 2024). Attribution and verification evaluate whether claims are supported (Wadden et al., 2020; DeYoung et al., 2020; Gao et al., 2023). LitTraceQA combines these concerns in one output contract: retrieve canonical papers, localize typed evidence, and return a checkable answer. Appendix C gives a component-level comparison.

5 Limitations and Conclusion

The current collection is concentrated in recent computer vision and NLP venues, and its source-access policy must respect the original publishers' terms. Evidence locators require further normalization across artifact types, and the official release should document public/hidden splits, licensing, expert review, and inter-annotator agreement where applicable. Closed-book challenge outcomes cannot substitute for retrieval-augmented baselines or human evaluation. These limitations define the remaining release work rather than evidence of completed system comparisons.

LitTraceQA reframes scientific QA as a traceable grounding problem. Its questions require systems to identify papers, locate heterogeneous evidence, and produce checkable answers; its evaluation makes each stage and their strict joint success visible. The 4,978-question collection provides a basis for studying where literature-grounded systems lose provenance between retrieval and response.

References

Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, Tanya Goyal, Danqi Chen, and Tianyu Gao. 2024. LitSearch: A retrieval benchmark for scientific literature search. *Preprint*, arXiv:2407.18940. Accepted to EMNLP 2024.

Akari Asai, Jacqueline He, Rulin Shao, Weijia Shi, Amanpreet Singh, Joseph Chee Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D'Arcy, David Wadden, Matt Latzke, Minyang Tian, Pan Ji, Shengyan Liu, Hao Tong, Bohao Wu, Yanyu Xiong, Luke Zettlemoyer, and 6 others. 2024. OpenScholar: Synthesizing scientific literature with retrieval-augmented lms. *Preprint*, arXiv:2411.14199.

Bernd Bohnet, Vinh Q. Tran, Pat Verga, Roee Aharoni, Daniel Andor, Livio Baldini Soares, Massimiliano Ciaramita, Jacob Eisenstein, Kuzman Ganchev, Jonathan Herzig, Kai Hui, Tom Kwiatkowski, Ji Ma, Jianmo Ni, Lierni Sestorain Saralegui, Tal Schuster, William W. Cohen, Michael Collins, Dipanjan Das, and 3 others. 2022. Attributed question answering: Evaluation and modeling for attributed large language models. *Preprint*, arXiv:2212.08037.

Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.

Jay DeYoung, Eric Lehman, Ben Nye, Iain J. Marshall, and Byron C. Wallace. 2020. Evidence inference 2.0: More data, better models. In *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing*.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Tiancheng Huang, Ruisheng Cao, Yuxin Zhang, Zhangyi Kang, Zijian Wang, Chenrun Wang, Yijie Luo, Hang Zheng, Lirong Qian, Lu Chen, and Kai Yu. 2025. AirQA: A comprehensive qa dataset for ai research with instance-level evaluation. *Preprint*, arXiv:2509.16952. Accepted to ICLR 2026.

Jakub Lála, Odhran O'Donoghue, Aleksandar Shtedritski, Sam Cox, Samuel G. Rodriques, and Andrew D. White. 2023. PaperQA: Retrieval-augmented generative agent for scientific research. *Preprint*, arXiv:2312.07559.

Yoonjoo Lee, Kyungjae Lee, Sunghyun Park, Dasol Hwang, Jaehyeon Kim, Hong-In Lee, and Moontae Lee. 2023. QASA: Advanced question answering on scientific articles. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 19036–19052. PMLR.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, pages 9459–9474.

Chuhan Li, Ziyao Shangguan, Yilun Zhao, Deyuan Li, Yixin Liu, and Arman Cohan. 2024. M3SciQA: A multi-modal multi-document scientific qa benchmark for evaluating foundation models. *Preprint*, arXiv:2411.04075.

Shraman Pramanick, Rama Chellappa, and Subhashini Venugopalan. 2024. Spiqa: A dataset for multimodal question answering on scientific papers. In *Advances in Neural Information Processing Systems*, volume 37, pages 118807–118833.

David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*.

Li S. Yifei, Allen Chang, Chaitanya Malaviya, and Mark Yatskar. 2025. ResearchQA: Evaluating scholarly question answering at scale across 75 fields with survey-mined questions and rubrics. *Preprint*, arXiv:2509.00496.

Xuanliang Zhang, Dingzirui Wang, Baoxin Wang, Longxu Dou, Xinyuan Lu, Keyan Xu, Dayong Wu, and Qingfu Zhu. 2025. SCITAT: A question answering benchmark for scientific tables and text covering diverse reasoning types. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3859–3881, Vienna, Austria. Association for Computational Linguistics.

A Detailed Output Contract

Each record contains a stable query ID, natural-language question, requested answer types, gold papers, typed evidence, and answer. Gold-paper entries store canonical IDs with title, venue, and year. Evidence entries store an evidence ID, supporting paper ID, source type, evidence text or value, locator, and grounding strings. Multiple-choice records include the option set. Structured tables include a schema and row-key columns used to align predictions with gold rows. The public development split contains 55 complete-schema examples: 26 hidden-source single-paper questions and 29 multi-paper questions, with all five primary evidence categories represented.

B Construction and Release Details

The source pool contains 27,487 metadata records from recent machine learning, computer vision, and natural language processing venues. A paper must have an inspectable local text cache to support generation. Open-book generation produces the question, answer, gold papers, and evidence together. Automated checks reject evidence attached to undeclared papers and test whether distinctive grounding strings occur in the source text. Memory-only challenge attempts and their errors are retained in the annotation to support later difficulty analysis. Corpus statistics count unique question text: records whose questions differ are distinct, while exact duplicate questions from cumulative checkpoints are removed.

The release should normalize locator fields before official scoring. Tables and figures naturally support page plus object ID; text spans, equations, algorithms, and citation contexts may additionally require section, region, equation ID, algorithm step, or cited-paper fields. Public distribution should include metadata and annotations but should not redistribute paper PDFs without permission. A release card should document source licenses, filtering, human review, model-assisted steps, known error rates, and the intended use of single- and multi-paper subsets.

C Comparison with Related Benchmarks

D Metric Definitions

For a question i, let P_i^* and $\hat{P}_i$ denote gold and predicted paper-ID sets. We macro-average set precision, recall, and F1 across questions. Ranked systems may additionally report Recall@K, MRR, MAP, or nDCG. Evidence items are normalized to tuples of paper ID, source type, page, and object ID when available. Coarse exact match requires the full tuple; evidence set F1 gives partial credit for recovering a subset of required items.

Multiple-choice accuracy compares the submitted option letter with the gold option. Structured-table rows are aligned by declared row keys; we report row F1 and both macro- and micro-averaged cell accuracy. Strict joint success for question i is

$$J_i = \mathbb{1}[\hat{P}_i = P_i^*]\,\mathbb{1}[\hat{E}_i = E_i^*]\,\mathbb{1}[\hat{a}_i = a_i^*], \quad (4)$$

with any component failure setting J_i to zero. Oracle-paper evaluation replaces $\hat{P}$ with P^*; oracle-evidence evaluation additionally supplies E^*. These are diagnostic conditions, not alternative task definitions.

E Representative Multi-Paper Annotations

Cross-paper robot manipulation comparison. One structured-table question asks for the real-world success-rate improvement and number of real-world tasks reported by FVP and Image-to-Sphere Policy. The gold trace uses Figure 1 context and abstract text from the FVP paper together with introduction text and the Figure 1 caption from Image-to-Sphere Policy. The answer is 28% over 12 tasks for FVP and 42.5% over 4 tasks for Image-to-Sphere Policy. All three closed-book challengers filled only one of five checked cells correctly. Their answers supplied plausible task counts or improvements, but failed to align each number with its source paper.

Cross-paper retrieval metric alignment. A second table question aligns headline gains from Diff2I2P, DAC, and BLiM+CPN across registration, open-set 3D object retrieval, and text–video retrieval. The gold evidence contains the result sentence from each paper plus an additional DAC context span. The answer records over 7% registration-recall gain on 7-Scenes, 10.01% mAP gain over four open-set 3DOR datasets, and a 6.4 average R@1 gain, respectively. Memory-only attempts produced smaller or mismatched improvements, illustrating a common failure: recalling a plausible metric without preserving the paper–dataset–value alignment.

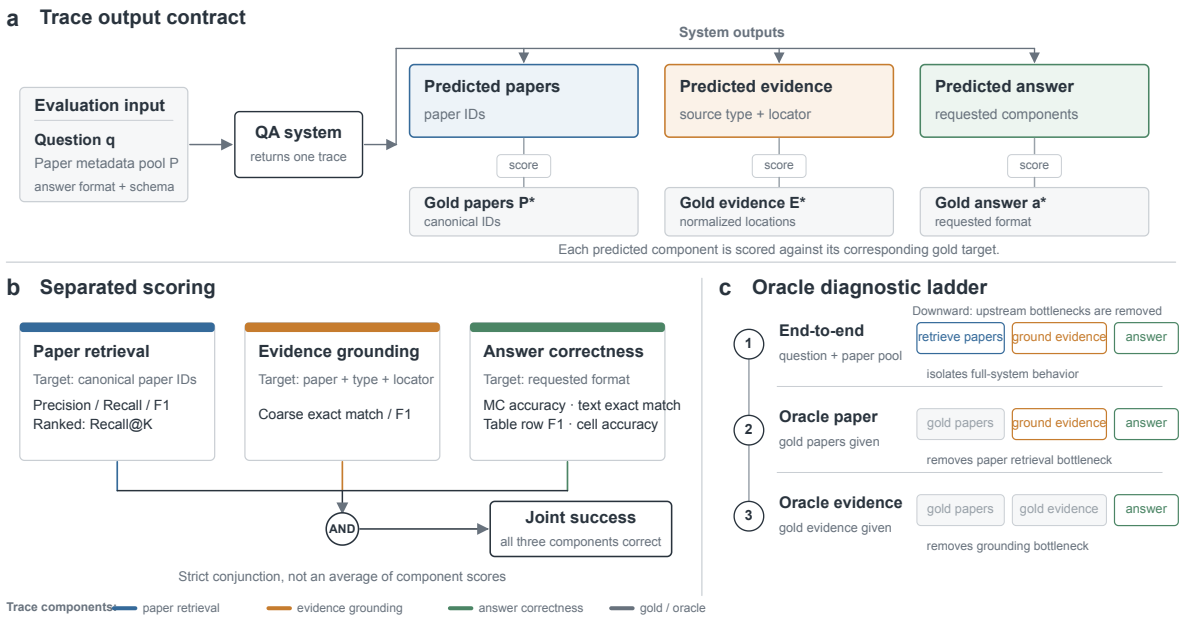


Figure 3: Detailed LitTraceQA evaluation protocol. Systems return papers, evidence locations, and answers. Component metrics diagnose each stage; strict joint success requires all stages to be correct; oracle settings remove upstream bottlenecks without implying additional baseline results.

Citation-context attribution. A multiple-choice question combines two LLM-compression papers and asks which cited works support an OBS-to-quantization adaptation and monotonic toxicity reduction under knowledge distillation. The gold option pairs Frantar and Alistarh (2022) with the former and Xu and Hu (2022) with the latter. All challengers selected a distractor that confused these cited sources with later GPTQ- or SparseGPT-style work. The error is not merely a wrong name: it breaks the link between the claim, its citation context, and the paper that makes the claim.

F Ethical Considerations

LitTraceQA evaluates access to and reasoning over scientific publications. A system may still reproduce errors, omissions, or biases in the source papers, and a traceable citation does not guarantee that the cited claim is correct. Users should inspect the supplied evidence rather than treat model answers as expert advice. Dataset documentation should identify model-assisted annotation, human review, source licensing, and any venue or field imbalance. The benchmark should not be used to redistribute copyrighted PDFs or to infer sensitive attributes about authors.

Table 1: Component-level comparison with scientific QA and retrieval benchmarks. A checkmark indicates an explicit prediction or evaluation target; a triangle indicates partial or implicit support.

Benchmark	Paper ret.	Multi-paper	Typed evid.	Checkable ans.	Strict trace	Hardness	Main distinction
QASPER	✗	✗	▲	✗	▲	✗	Given-paper QA with supporting spans.
QASA	✗	✗	▲	✗	▲	✗	Full-paper reasoning without pool retrieval.
M3SciQA	▲	✓	✓	✗	▲	✗	Multimodal multi-document scientific QA.
SPIQA	✗	✗	✓	✗	▲	✗	Figure/table QA with source context supplied.
SCITAT	✗	✗	✓	✓	▲	✗	Table-text reasoning rather than paper retrieval.
AirQA	✓	✓	✓	▲	▲	✗	Broad academic QA settings and evaluators.
LitSearch	✓	✗	✗	✗	✗	✗	Literature retrieval without grounded answers.
ResearchQA	▲	✓	▲	✗	✗	✗	Long-form scholarly QA with rubric scoring.
LitTraceQA	✓	✓	✓	✓	✓	✓	IDs, typed evidence, and answer are jointly required.